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OPERATIONS STRATEGY AT GALANZ

Dr. Stephen Ng and Barbara Li wrote this case under the supervision of Professors Xiande Zhao, Xuejun Xu and Yang Lei solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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BACKGROUND

On January 4, 2004, Liang Zhaoxian, the vice-chairman and chief executive officer (CEO) of Galanz Enterprises Group Co. Ltd. (Galanz), had just returned to his office after signing a contract to outsource part of the production of Galanz's designed and branded magnetrons[1] to a Japanese manufacturer. Galanz had previously relied on the magnetrons designed and branded by its suppliers during its earlier years of development as a microwave oven manufacturer for many foreign brands.

In the past, Galanz had purchased all its magnetrons (a core component of the microwave oven) from foreign suppliers, such as Toshiba and Panasonic. But the rapid growth of Galanz's microwave oven business had threatened these magnetron suppliers, who were also in the microwave oven market. In an effort to restrain their competition, these suppliers decided to reduce the magnetron supply to Galanz. This decision prompted Galanz to initiate a major investment in magnetron R&D in 1997. Eventually, the company was able to design and produce its own magnetrons to support its microwave oven production in 2000.

However, due to the tremendous growth in its microwave oven business, Galanz was still short of the component. By the end of 2003, its magnetron factory had an annual production capacity of 16 million units or 67 per cent of the total production requirement of 25 million units. Galanz found itself having to outsource part of the magnetron production to other original equipment manufacturers (OEMs), since by then its customers had begun to insist that Galanz use its own branded magnetrons in Galanz-supplied microwave ovens.

Although Liang was relieved that he had solved the problem of the magnetron supply for the company with the outsourcing agreement he signed with the Japanese manufacturer, he could not stop wondering what he should do to further guide the company to greater future success in the increasingly competitive

[1] *Magnetrons are high-powered vacuum tubes that generate the microwaves used in microwave ovens. For more details, see www.gallawa.com/microtech/magnetron.html.*

market. Liang knew that this interesting dilemma was the result of the company's continuous growth and transformation.

In the early days, Galanz only produced microwave ovens for the domestic (China) market with its own brand, while the production technology and key component parts were purchased from Japan. It then started its OEM business when foreign brand owners outsourced production to Galanz due to its very low cost. When the company gradually mastered the design and production of magnetrons, Galanz transformed itself from an OEM to an original design manufacturing (ODM) firm. Since then, the company had operated in the original brand manufacturing (OBM) mode in the Chinese market and a combination of the OEM and ODM modes in the overseas market. In recent years, due to increasing recognition of its brand, its OBM business had begun to experience growth in the overseas market.

Galanz's capability of producing microwave ovens at a low cost, combined with its enhanced R&D ability, had allowed it to compete successfully with major players such as Panasonic, Toshiba and LG in the global electrical appliance market. Galanz had gained a leading position with more than 50 per cent of the global market share in microwave ovens in 2007. Its brand name was well known in China and overseas. Products in both OEM and OBM versions were sold in the U.S., European, South American and African markets. The total sales volume of all types of Galanz microwave ovens climbed from two million units in 1997 to 22 million units in 2005. Its revenue increased from RMB2.96 billion in 1999 to RMB18 billion in 2006 (see Exhibits 1 and 2 for figures on production, sales, revenue and profit for Galanz microwave ovens).

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GALANZ

Galanz, headquartered in Shunde, Guangdong province, was originally a township enterprise employing a few workers dealing in the trading of down feather products. It was founded in 1978 by Liang Zhaoxian's father, Liang Qingde (Liang Senior hereafter), the former deputy chairman of the Industrial and Transportation Office of Shunde County (1973-1978). The company's original name was Guizhou Down Product Factory. It produced down feather products for overseas clients to earn foreign exchange.[2] At that time, companies were not permitted to export products without a quota. Since the company was jointly owned by the Foreign Trade Department of Guangdong province and the Shunde government, it was able to obtain the required quota and by 1992, its export volume was more than RMB 23 million.

Although the company continued to be profitable despite increasing competition, the global garment industry faced dramatic infrastructural changes that posed potential risks to its business in the late 1980s. The export quota also restricted the growth of the company. Even though the premier of the State Council, Zhao Ziyang, visited Liang Senior in 1988 and promised to strategically develop the industry by freeing it from further trading tariffs, the leader's proposal eventually proved ineffective and the industry's marginal profit continued to decline. Although Guizhou Down Product Factory enjoyed revenue of more than RMB 100 million at that time and was one of the best performers in Shunde, the profit growth had become stagnant and Liang Senior realized that his business could not go any further because of its projected weak future. Thus in 1991, Liang Senior made a strategic decision to search for new business opportunities with greater potential for growth. After analyzing the Chinese consumer market for a year, he made the decision to enter the electrical appliance market with the introduction of the microwave oven.

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[2] Industrialization in China started with the economic reforms of 1978. At the time, non-state enterprises such as Galanz began to engage in production known as "sanlai yibu" (meaning three resources and one obligation), in which products were manufactured with imported materials, processed according to foreign models, and assembled according to foreign clients' requests.

Another reason for the transition of the company was the change in the institutional environment in the 1990s. For reasons of ideological acceptability, Liang Senior could only register the company as a collective enterprise[3] jointly owned by the Foreign Trade Department of Guangdong province and the Shunde government, with himself as general manager. This gave the company the privilege to not only obtain an export quota but also to access public funds and thus resources for development. Later, the transformation of the centrally planned economic environment was further energized when the retired Communist Party leader Deng Xiaoping made his rounds to southern China in 1992, emphasizing the further economic construction of the country by first opening up Guangdong and Shanghai as industrialized areas. Consequently, individual entrepreneurs were free from state control. Since then, the Shunde local government had begun to offer more legal recognition and freedom to privately owned enterprises by protecting the property rights of production assets, innovation and capital, and attracting foreign investments and industrial manufacturing development in the area. These private ownership rights enabled the company to grow in a stable institutional environment. At this juncture, Liang Senior embarked upon his new business of microwave ovens in 1992.

In 1994, a rare flood engulfed the Pearl River Delta, and Guangdong became the most serious disaster area. The flood submerged the entire production factory and Liang Senior seized the opportunity to buy all the shares from the two government shareholders. After allocating a portion of the shares to his comrades, Liang Senior then became the major shareholder of the company. In 1999, Liang Senior closed his down product factory, thus marking the completion of Galanz's transition to the microwave oven industry.

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WHY MICROWAVE OVENS?

On a visit to Japan in 1991, Liang Senior identified the microwave oven as a product with great potential in China. During that time, microwave ovens were all imported and sold at relatively high prices that made them unaffordable to most Chinese households. Liang Senior observed that the rise of the Chinese economy would stimulate purchasing power and the demand for a wide range of modern commodities in the country. China's modernization process was changing the living styles and habits of the Chinese people and attracting them to time-saving conveniences, including new ways of cooking and food preparation, so that microwave ovens had the potential to become popular and indispensible in modern cities in China. Liang Senior recognized that if he could produce and sell them at an affordable price, this would be an excellent business opportunity.

From the perspective of the competitive environment, the microwave oven market in the 1980s was in its infancy in China, where competition and demand size were small. Only a few foreign brands such as Toshiba, LG and Whirlpool were in the market and they had no clear intention to expand and dominate because they had not yet sensed the market potential, due in part to their lack of familiarity with the rapidly evolving Chinese market environment. Besides, the high price of their microwave ovens was unaffordable to most Chinese consumers.

On the technology side, it was less risky to invest in the technology associated with microwave ovens because this technology had been mature and stable since Americans first invented the microwave oven in the 1950s. Though Japan, a new player in microwave ovens, later advanced the production of the magnetron tube and power supply in the subsequent decade, resulting in a tremendous cost reduction,[4] the overall technology of microwave ovens did not differ much from that of its original design.

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[3] "A Fresh Look at the Development of a Market Economy in China," China Perspectives, July-August 2003, http://chinaperspectives.revues.org/document564.html, accessed February 25, 2009.
[4] "The Greatest Discovery Since Fire," American Heritage,

THE START-UP

Despite the technical ease of producing microwave ovens, starting a microwave oven production business in China was not without obstacles due to the lack of associated technology and technical expertise at home. While many Chinese entrepreneurs at the time saw the market opportunity in microwave oven production in China, only Liang Senior had the determination to work to overcome the business challenges.

Liang Senior understood that Galanz had to import equipment and technology from overseas. Thus, in the early 1990s, he purchased the microwave oven production blueprint for USD$300,000 from Toshiba, then the world leader in microwave oven production equipment and technology. He also searched for engineering professionals throughout the country to set up the factory. At last, he found a group of engineers from Shanghai No. 8 Radio Factory who were knowledgeable of microwave oven technologies to help him. These engineers were still working with Galanz after 25 years, serving in senior positions in the technical supervision of microwave oven production.

In 1992, Galanz produced its first microwave oven, the factory opened and the company officially changed its name to Guangdong Galanz Enterprises Group Co. Ltd. In 1993, the first batch of 10,000 microwave ovens was produced. In 1995, Galanz sold 250,000 microwave ovens in China, representing 25.1 per cent of the domestic microwave oven market, and overcame Shell Electric (later acquired by Whirlpool) as the leading domestic microwave oven manufacturer. Exhibit 3 lists the course of development of Galanz's microwave oven business.

EARLY SUCCESS

At the start, Galanz did not have any competitive edge in production technology but only an abundant supply of cheap labor[5] and land.[6] Offering a low price was thus the only way to compete in the market. From 1996, Galanz adopted a low-price strategy by repeatedly implementing cycles of price cutting and production capacity expansion. In 2003, the annual market demand for microwave ovens surged to 25 million. This remarkable achievement was built upon Galanz's persistence with its low-cost strategy, which relied on two tactics (explained below) by which Galanz was able to occupy the market-dominating position for years. In fact, the strategy was so successful that its production demand for microwave ovens grew faster than its magnetron production capacity (see Exhibit 3). The internal production capacity of magnetrons was limited to 16 million units, but the annual demand was 25 million units in 2003. The company therefore decided to outsource the magnetron production to a Japanese supplier for the balance of nine million units.

Transfer of Production Line Through OEM Agreement

To further achieve a low cost of production, the first tactic that Galanz used was to escalate its production capacity through a free production line transfer. That is, when Galanz produced microwave ovens for Fillony, its French customer, it proposed to supply Fillony with the necessary quantity of microwave ovens at cost if Fillony agreed to transfer, for free, its entire production line as well as the assembly technology to

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www.americanheritage.com/articles/magazine/it/2005/4/2005_4_48.shtml, accessed February 25, 2009.

[5] In 1997, the average income in China was RMB494.42 per month, and in the United States it was $553.14 per week. http://laborsta.ilo.org, accessed February 25, 2009.

[6] Liang Senior was a member of the Communist Party of China (CPC). With his political background and guanxi (good relationships) with the local government, Galanz could access cheap land and local resources.

Galanz, and to provide the necessary training to Galanz's operators. More importantly, Galanz gained the right to use the excess capacity of the production line for its own products after satisfying Fillony's production quota. Galanz continued to apply this business tactic to other big appliance labels, including Toshiba, Sanyo, Whirlpool, General Electric (GE), SEB, and DeLonghi, which were willing to provide staff training, parts customization, and production site improvements that allowed Galanz to quickly enlarge its production scale and escalate the product quality.

More interestingly, the same tactic was applied to attract component suppliers setting up component production facilities in Galanz. For example, Galanz offered a deal to a European transformer supplier who originally produced transformers in Europe at a cost of $30 each. If its production facilities were transferred to Galanz, Galanz committed to supply the necessary units for $8 each exclusively to this supplier but owned the right to use the surplus capacity after fulfilling the orders. A Japanese company that produced transformers at $20 each was also attracted by such a low-price offer and agreed to the same arrangement when proposed by Galanz. As a result, Galanz committed to produce transformers for this Japan supplier at $5 each and obtained its production line free.

To further increase its production capacity, Galanz fully utilized its production facilities and labor resources to the extreme by operating three shifts per day, seven days a week and 365 days a year. Compared with many Western companies that operated production only 30 to 40 hours per week, Galanz's non-stop production had expanded its production capacity to at least four times that of its Western counterparts. In sum, this tactic allowed the company to increase its production scale and to reduce production costs to a level that Galanz's competitors could not beat.

Price War

After Galanz proved able to sustain the cost leadership, the second tactic was to repeatedly launch a price war such that it could fully dominate the domestic microwave oven market (see Exhibit 4 for an illustration of the price-cutting cycle). When international brands first entered the Chinese market in the early 1990s, their products were priced at a level that was not affordable for the average consumer in China. Prices of microwave ovens ranged from RMB1,000 to RMB3,000. Consequently, Galanz decided to cut prices to gain market share from its competitors. From 1996 to 1998, Galanz cut prices of all its microwave ovens by 30-40 per cent. Its lowest-priced microwave oven could be less than RMB 300. It continued to launch price wars between 2000 and 2002 and finally the company achieved its leading position in both the domestic and global markets (see Exhibits 3 and 5 for stages of Galanz's price war).

Galanz's price-cutting mechanism relied on setting the average unit cost of production along its growth curve. That is, when production volume reached two million units, the price was set based on the average cost of producing 0.8 million units. When production volume reached eight million units, the price was set based on the average cost of producing five million units. When production reached 15 million units, the price set was based on the average cost of producing 12 million units, and so on.

In 1996, Galanz's production volume was 4.5 million units, but by the end of 2001, it jumped to 30 million units. At that stage, Galanz deliberately expanded its production capacity of microwave ovens to exceed the market demand. Since it enjoyed tremendous economies of scale for low-cost production, the company pushed its sales team to work harder so that the market could absorb the additional inventory. Galanz's aggressive pricing strategy led many industry players to withdraw from the market.

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After seven rounds of price cuts, Galanz had dramatically increased its domestic sales in 2002. In 1997, Galanz occupied 47.6 per cent of the domestic market with sales reaching 1.25 million units, which was five times the sales in 1996. By the end of 2002, its domestic sales were four million units, occupying 70 per cent of the domestic market. As a result, LG and Panasonic retreated from the Chinese microwave oven market because of Galanz's dominant market position. As Liang commented, "The main objective of the price war was to destroy our competitors' confidence to compete with us when they realized that the market bore little investment value. The price war has been a line of defense for us." Once at the top, Galanz began to adopt a new pricing strategy by pricing products based on the extent of market acceptance. Targets for product pricing and profit margins were determined by counting backwards. This strategy drove a series of efficiency improvement projects to cater to the predetermined business targets.

TRANSFORMATION FROM OEM TO ODM

While the launch of price wars had stimulated the domestic demand for microwave ovens, Galanz began to face a shortage of magnetrons because of the retrenchment by its suppliers. Toshiba Hokuto Electronics had been Galanz's major magnetron supplier since 1993. The exponential growth of Galanz in the early years created an internal shortage of the component within Toshiba, and this seriously compromised Toshiba's microwave oven business. Toshiba's headquarters in Japan thus requested that its subsidiary stopped supplying magnetrons to Galanz. Galanz had to ask Panasonic for help. But its board of directors also planned to set up microwave oven production lines in Shanghai for the Chinese market. After negotiations with these suppliers, Galanz was only able to get commitments for three million units of magnetrons per year from each. These numbers were far below Galanz's requirements. LG and Daewoo then agreed to supply Galanz, but later also refused for similar reasons.

Development and Production of Own Magnetron

This supply crisis motivated Galanz to design and develop its own magnetrons, and to switch its business direction from "Made in China" to "Created in China." Galanz felt that the technical capability established so far, which was based on learning and importing the most advanced technologies from overseas partners, had equipped the company with the necessary technical know-how to design and develop its own magnetrons.

Since 1997, Galanz had collaborated with South China University of Technology, the Chinese Academy of Sciences and Guangdong Academy of Sciences to develop its own magnetrons. In 2000, the research team successfully launched the first Galanz-made magnetron with an improved mechanical design to enhance quality and function compared to the Japanese magnetrons. The team also brought technological breakthroughs in microwave cavity matching and micro switching. Subsequently, Galanz formed Galanz Magnetron Subsidiary Company to begin mass production (see Exhibit 6 for samples of magnetrons). The first-year production in 2000 was 10,000 units per day, increasing to around 44,000 units per day or 16 million units annually by 2003.

Enhancement of R&D Capability and Production Innovation

After seeing the benefits gained from its own R&D activities in the design and production of magnetrons, Galanz began to increase its investment in R&D in order to enhance its internal R&D structure and facilitate new product design and development. In 1995, Galanz established the Research Institute of

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Household Electrical Appliances in the Chinese headquarters and in 1997 established the Galanz American Research Center in the United States.

Its investment in R&D represented more than three per cent of annual revenue. After mastering magnetron technology, Galanz carried on product innovations by focusing on developing new features and new technologies. As a result, it changed its cavity-matching design; improved its power supply specifications; added new cooking functions, such as steaming, grilling, boiling and stewing; improved existing product structures, such as express cooking and intelligent LCD menu touchpads; and redesigned the product appearance with titanium-film mirrors and edgeless flat organic glass. The new technological invention of light wave ovens in 2001 also lifted Galanz into the high-end product market.

The technical capability of Galanz, established through both investment in R&D and through learning and importing the most advanced technologies from overseas partners, allowed the company to reduce costs and provide differentiated features in its products. Meanwhile, the company started providing more ODM service to its large OEM clients as well as receiving more orders for products with the Galanz brand from small and mid-size enterprises in developing countries. As the marketing manager noted, "In the old days, we only followed the trend. But now, we realize the importance of innovation. We now strive to localize our design to suit each target market."

The Drive for Self-Sufficiency

As an additional effort to reduce costs and improve quality, Galanz started to vertically integrate its supply chain by manufacturing more of its components. About 90 per cent of the microwave oven parts were produced by Galanz. In fact, for magnetrons and associated core components, Galanz developed the manufacturing technologies on its own. But for non-core components such as transformers, Galanz would work with suppliers to transfer their production lines and assembly technology to the company. However, since Galanz's production demand was so large, materials and some parts had to be sourced from external suppliers. About 3,000 suppliers were engaged, with 60 per cent located in Shunde, 30 per cent from nearby cities in the Pearl River Delta Region and 10 per cent from other areas in China.

OEM/ODM versus OBM in the Overseas Market

Galanz stayed on top in the domestic microwave oven market. Its production quality earned acclaim among international industry players, but overseas consumers were not familiar with Galanz as a brand of microwave ovens. By 2003, Galanz's primary exports were OEM microwave ovens, which had no brand recognition to end consumers. Strategic partnerships with multinational companies (MNCs), such as K-Mart and Wal-Mart, were confined to OEM deals.

Liang Junior, however, began to experience a change in the strategic relationship with these MNCs. First, he saw that globalization was causing fierce competition among these MNCs, which pushed them to reach for new markets and branded products. Second, he was aware that his products were widely recognized for their low cost and good quality. As a result, he began to offer Galanz-branded microwave ovens to K-Mart and Wal-Mart superstores. Another example of Galanz's OBM effort was the technical support from Fillony to help Galanz set up R&D centers overseas to showcase Galanz-branded microwave ovens. In Indonesia, Cosmos, the largest consumer goods distributor, asked to be the sole agent of Galanz-branded microwave ovens in the country.

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Through concerted efforts by Galanz and its partners, the ratio of Galanz OBM and OEM microwave ovens rose from 1:9 to 3:7 from 1997 to 2003. However, maintaining an OBM and OEM strategic position was easier said than done. Galanz faced the problem of low brand awareness by overseas consumers when it launched its branded products overseas. For instance, when Galanz promoted seven of its microwave oven models in K-Mart and Wal-Mart, none of these model ovens was recognized by overseas consumers, resulting in very poor sales performance. Finding a way to increase overseas consumers' awareness of Galanz's brand imposed challenges and pressures on the company.

Galanz was also facing accusations from the governments of importing countries. The company was being accused of monopolizing the market by dumping products with prices set unreasonably low. For example, the Argentinean government launched an anti-monopoly lawsuit against Galanz when its OBM microwave ovens reached a market share of 70 per cent.

To further expand OBM sales, Liang knew that the company had to set up sales and service networks in the target markets either on its own or through its strategic partners. Furthermore, as the company expanded its OBM sales, Galanz was concerned that it might gradually become a competitor of its OEM customers. Liang sensed that some of Galanz's OEM customers had felt threatened by the situation and that they might cut their OEM orders from Galanz. Therefore, Liang felt that Galanz must be careful not to harm the interests of its OEM customers in its effort to expand its OBM sales if it still wanted to maintain the OEM business.

ORGANIZATIONAL STRUCTURE AND SYSTEMS

It took Galanz about a decade (1993-2003) to enter the global home appliance market. Then it became the dominant player in the world microwave oven market. At this point, Galanz made the decision to transform itself from being the "World Factory" to being the "World Brand." Liang realized that this new strategic direction created unprecedented challenges for the various aspects of the business.

Management Structure

Led by Liang Junior and Liang Senior, the company adopted a highly centralized decision-making and execution system that only had three levels of personnel: senior directors, general managers and operations staff (see Exhibit 7 for the structure in 2002). Information flow and communication were slow and unclear. Strategies were difficult to implement. As the company grew from 20 employees in 1993 to more than 10,000 employees in 2003, the management structure that had served Liang well in the past became inadequate for the rapidly changing competitive environment. In fact, when Liang Senior reviewed the organization structure during this period, he commented that, "Our management structure has been highly centralized and it has been suitable during our development and growth. When Galanz was a small factory, it was easy for us to master every detail. But now, when we have thousands of employees, can we still rely on the perspective of a few people for every decision?"

Manufacturing System

Microwave oven production continued to expand, with the company going from having one production line with 300-500 output units per day in 1993 to 14 production lines each with 1,100 output units per day in 1998. Later, by integrating component production and the assembly process, Galanz developed 24

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production lines with 50,000 gross output units per day by running three shifts and operating 24 hours a day. However, the increasing demands of customized products had caused a decline in production efficiency. Liang wished that production systems and facilities were designed to manage product diversity, particularly for the more diversified OBM business.

Liang also recognized the capacity problem with its own magnetron production, which was unable to cope with the ever-increasing demand for microwave ovens. He had to find a long-term solution to narrow this production gap. Though Galanz had outsourced part of its magnetron production for now, Liang was still concerned with ensuring the availability and the high quality of this key microwave oven component.

He also knew that the company overlooked the importance of data records and compliance with codes of practice that could help optimize its production efficiency. As new facilities and systems came online, employees had to be well-trained to use and repair the machinery. Liang realized that he should not solely rely on low-cost labor but also on workers' technical and management skills, work attitudes and efficiency.

Production Planning

Galanz had long adopted large-scale production to achieve low-cost efficiency, and in the past, most products were low quality with less variety. Sales turnover was high because of the large demand for low-price microwave ovens. In fact, when Galanz had a large production scale with low product variety in the domestic OBM market, it used to push its products to market based on its own forecasts. It strived to produce more than the forecasted demand, as it was confident that the extra inventory would eventually be absorbed in the seller's market. Therefore, little effort had been put into strengthening the capability of sales forecasting and production planning.

Zhao Jing, manager of sales and marketing, commented, "Inventory motivates sales people to work harder and make more sales." While this may have worked effectively during earlier years when the products were more standardized and design changed less frequently, the situation was entirely different by 2003. First, the overseas OBM market was often characterized by small and varied demand with high product variety. As Galanz developed a full range of low- to high-end products with more complicated product configurations, the company needed to be able to forecast the demand accurately and produce the right products in the right quantity at the right time to meet the needs of its customers. If the company failed to do so, it would be faced with unsold inventory for certain products while suffering from a shortage of products that the consumers wanted.

In a purely OEM business, the customer placed the order and paid the company on delivery. In this mode of operation, the customer bore the risk of poor forecasting and planning. In the OBM business, however, Galanz had to bear these risks. Therefore, the accuracy of the forecast and the responsiveness of its planning process played key roles in the company's success.

In addition, due to the shortened product life cycle and increased product variety, customers often requested a variety of different product configurations and ordered smaller quantities of each item. The traditional push mode of production could lead to overstock of unpopular products and shortages of popular products. Therefore, the question became, How does Galanz adapt the mass production system to meet the needs of low volume and high variety products as demanded by customers? Liang recognized that the answer to this question was essential to the success of the OBM business in the overseas market.

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On top of these issues, it was also challenging to decide which production orders should receive higher priorities when OEM, ODM and OBM orders were all waiting to be fulfilled, especially when the company was faced with capacity limits or material shortages.

Synchronization Problems

Galanz took a great leap to establish technological know-how early on, but was not immune to difficulties in maintaining a high quality of technological output. For example, the material detection testing program had to improve to minimize product irregularity, the time-consuming development of steel die delayed the progress of R&D, and because of Galanz's 24/7 production schedule, high-output machinery experienced rapid wear and tear, necessitating frequent repairs.

Synchronizing production with the R&D department became a challenge. The production department often failed to arrange prototype production and testing, thus delaying the new product development process. For instance, the slow progress in solving the mismatching problem between the design of an oven structure and cavity was always due to the misalignment in scheduling between design and production, as well as a communication gap. This kind of internal conflict had existed between the R&D and production departments during the large-scale mass production era of the past, but it was now exacerbated with the small-scale production of customized products.

Marketing Channel and Customer Relationship Management

Galanz had been strengthening its sales network in all parts of the world, including China, during recent years. There were 52 sales offices across China and branches were set up in major exporting countries. Compared to its competitors with long experience in managing international sales networks, Galanz's existing sales infrastructure and expertise were, however, not substantial enough to help its OBM products reach end-user markets, especially when its competitors had already adopted global marketing strategies and provided after-sales support worldwide. Their global networks were substantiated by their intensive capital investment, but Galanz was just beginning to put in the resources to build up exposure in the global market. Cultural differences in overseas markets and getting market information know-how hampered the success of popularizing Galanz with global consumers.

As Galanz decided to launch its branded products in the overseas market, the company had to provide a competitive level of service to a large pool of end customers with diverse needs and expectations. In the past, questions did not reach the company directly because its OEM clients had already handled them for Galanz. Customer support including after-sales services, product repair and maintenance, provision of product-safety knowledge, handling of customer complaints and returns, and warranty claims required significant work on Galanz's part to train the corresponding staff and establish a communication platform to handle end-users' requests. In sum, the company had to invest resources to enhance its customer service capabilities.

THE FUTURE OF THE COMPANY

Despite Galanz's spectacular growth, the company was facing many challenges. Liang wondered whether the low-cost competitive strategy that had been used successfully for years was still effective when Galanz's products and role in the home appliance market experienced critical changes. Liang realized that he had to address several important issues that could influence his company's strategic direction and

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success. Specifically, how should Galanz align its competitive strategy with the current operations configuration characterized by the combination of OEM, OBM and ODM? Should Galanz continue its pervasive low-cost strategy? Would the development and growth of the OBM business in the overseas market negatively influence the OEM business? And if so, how should Liang respond? How could the company's R&D, production and marketing functions effectively support the requirements of OEM, OBM and ODM customers? How should the company set priorities and utilize its resources and capabilities to gain competitive advantages in the marketplace? Should Galanz continue to expand its production capacity of magnetrons through outsourcing? What would be the impact of this in the long run? All these questions bombarded Liang's mind. He needed to formulate a new operations strategy to guide Galanz to even greater success in the dynamic global marketplace.

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Exhibit 1

PRODUCTION AND SALES OF GALANZ MICROWAVE OVENS (1992-2003)

Year	Sales Units (Million)				Market Share %	
	Domestic (OBM)	Export (OEM)	Export (OBM)	Total	Domestic	International
1997	1.25	0.07	0.68	2.00	47.60	‹10.00
1998	3.15	0.40	0.95	4.50	61.40	15.00
1999	3.00	1.50	1.50	6.00	67.10	25.00
2000	4.00	3.60	2.40	10.00	76.00	30.00
2001	6.00	3.00	3.00	12.00	70.00	35.00
2002	4.00	6.23	2.77	13.00	70.00	40.00
2003	5.00	7.54	3.46	16.00	60.00	44.50

Source: Galanz Enterprises Group Co. Ltd.

Exhibit 2

REVENUE AND PROFIT OF GALANZ MICROWAVE OVENS (2000-2003)

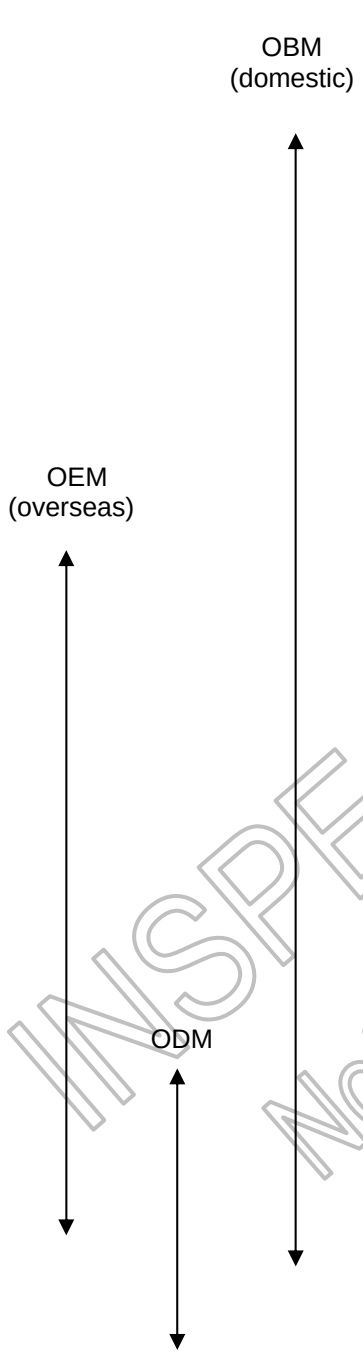
Year	Revenue (RMB Million)	Profit (RMB Million)
2000	5,600	360
2001	6,800	330
2002	9,000	320
2003	10,100	480

Source: Galanz Enterprises Group Co. Ltd.

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Exhibit 3

TRANSFORMATION OF GALANZ GROUP IN MICROWAVE OVEN BUSINESS



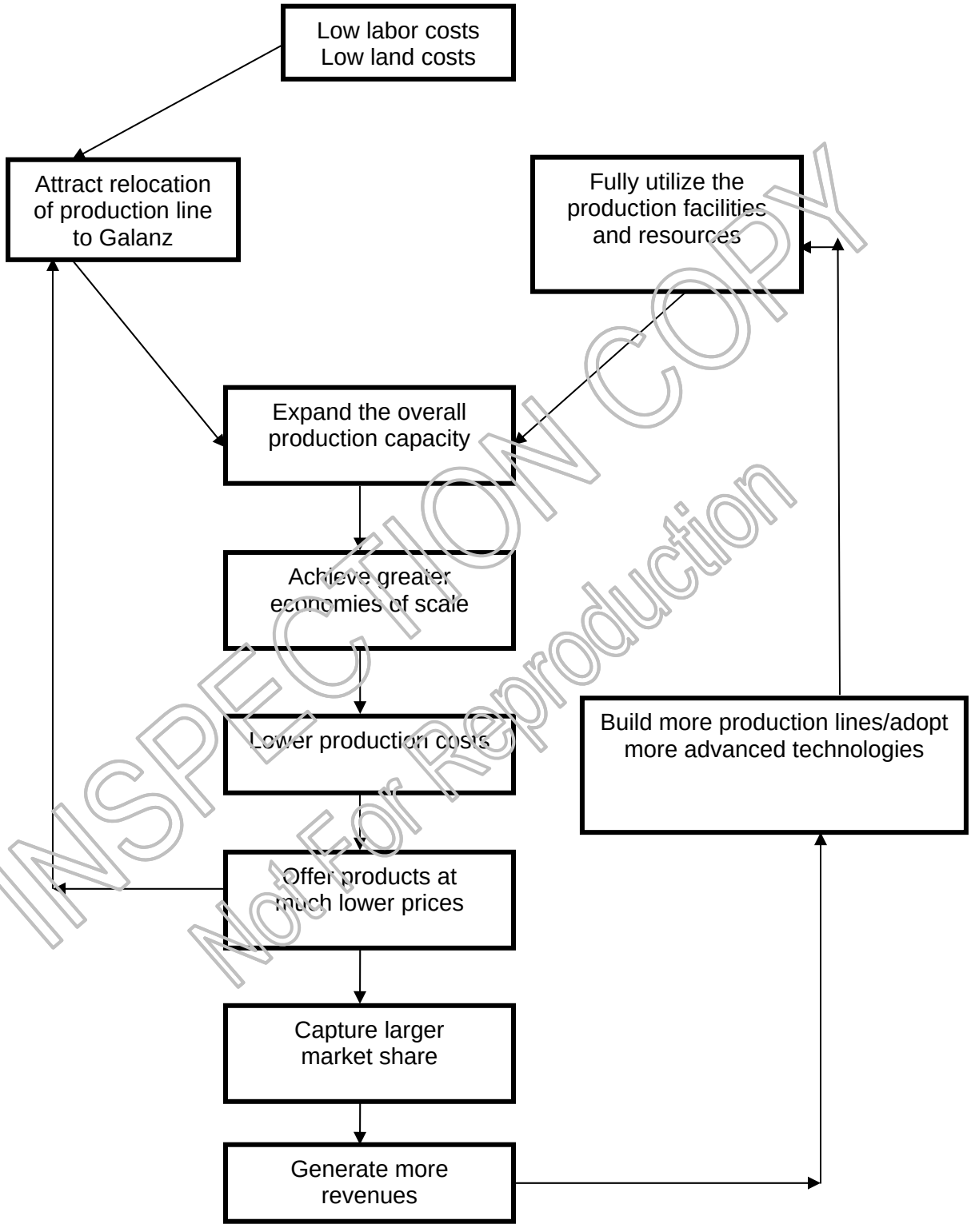
Year	Milestones
1991	• Bought the blueprints and production lines of Toshiba microwave ovens for $300,000. • Sought assistance of engineers from Shanghai No. 8 Radio Factory to build factory in Shunde.
1992	• First Galanz microwave oven. • Name changed to Guangdong Galanz Enterprises Group Co.
1993	• Produced 10,000 microwave ovens for trial sales.
1994	• Microwave oven production increased to 100,000 units.
1995	• Sold 250,000 microwave ovens, occupying 25.1% of the domestic market. • Replaced Shell Electric as the leading microwave oven manufacturer in China. • Established Research Institute of Household Electrical Appliances in China.
1996	• Received the first OEM order and started export sales. • Panasonic and Toshiba limited the magnetron supply to Galanz. • Galanz started a six-year price war lasting until 2002.
1997	• Total annual sales reached 2,000,000 units. • Domestic market share was 47.6%. • Won the national title of No. 1 brand of microwave oven in China. • Started developing own magnetron. • Established Galanz American Research Center in the U.S.
1998	• Annual production output reached 4,000,000 units. • Obtained product certifications from major European countries. • Built the world's largest single facility in microwave oven production.
1999	• Officially closed down the feather garment factory. • Set up R&D centre in the U.S. • Set up sales subsidiaries in Canada and the U.S.
2000	• Annual sales reached 10,000,000 units. • Domestic market share was 76%. • Succeeded in developing own magnetron. • Operated OEM and ODM production.
2001	• Epochal innovation of digital light wave ovens first appeared.
2002	• Annual sales reached 13,000,000 units. • International market share was 40%. • Price war ended. Samsung and LG retreated from the domestic microwave oven market.
2003	• All OBM microwave ovens used self-developed magnetrons. • Annual production of magnetrons reached 16,000,000. • Brand received increased recognition in overseas market.

Source: Galanz Enterprises Group Co. Ltd.

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Exhibit 4

GALANZ PRICE-CUTTING CYCLE



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Source: Galanz Enterprises Group Co. Ltd.

Exhibit 5

STAGES OF GALANZ PRICE WAR IN CHINA (1996-2002)

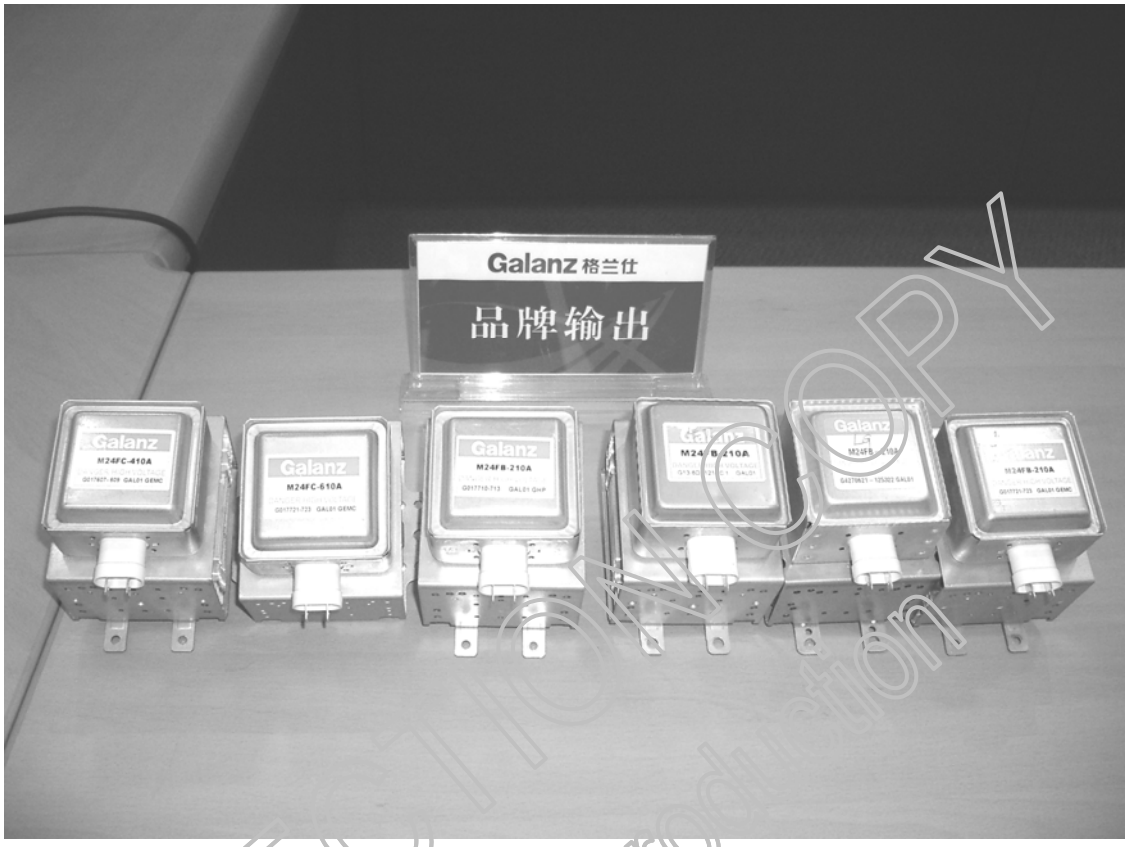
Month/Year	Product	Price Reduction	Result
Aug. 1996	Full range	40% discount	• Sales rose to 650,000 • Domestic market share more than 35%
Oct. 1997	Full range	29-40% discount	• Sales rose to 1,980,000 • Domestic market share more than 47.6%
May 1998	Full range	30% discount	• Sales rose to 4,500,000 • Domestic market share more than 60%
June 2000	Fine Golden Flower Series (mid-ranged products)	40% discount	• Sales rose to 1,000,000 in the two product lines • Domestic market share more than 76%
Oct. 2000	Black Edition Series (high-end products)	40% discount	• International market share more than 30%
Apr. 2001	Products below RMB 300	30% discount	• Products were popular in the low season
Jan. 2002	Digital Temperature Control Series	30% discount	• Galanz dominated the microwave oven market

Source: Galanz Enterprises Group Co. Ltd.

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Exhibit 6

MAGNETRONS DEVELOPED AND PRODUCED BY GALANZ (2000-PRESENT)

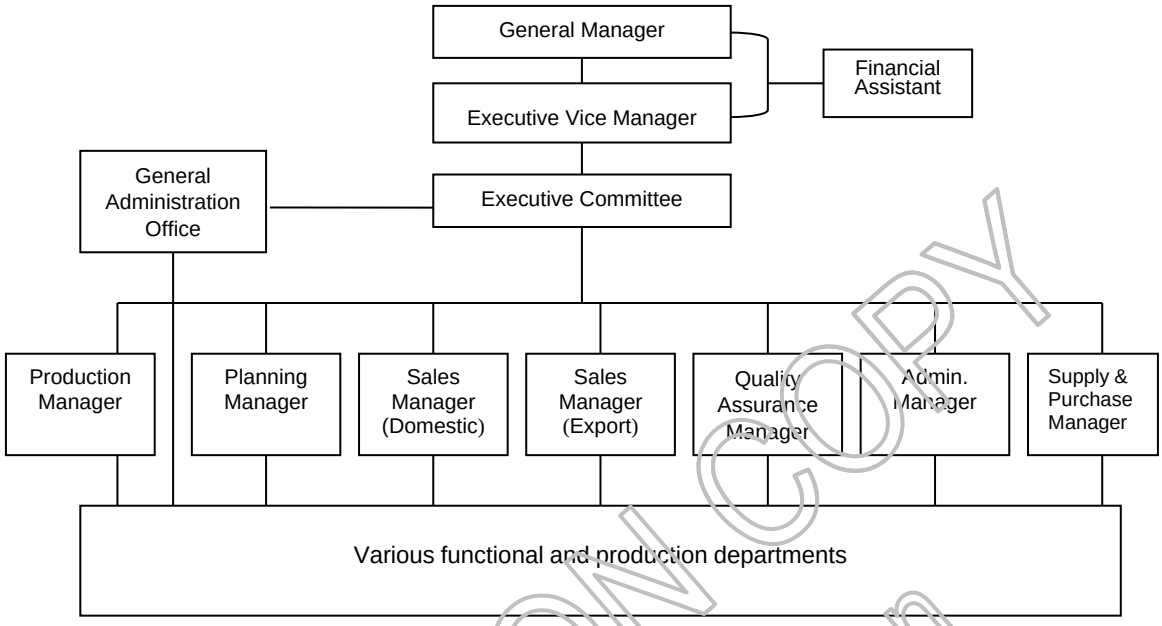


Source: Galanz Enterprises Group Co. Ltd.

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Exhibit 7

COMPANY STRUCTURE OF GALANZ GROUP IN 2002



Source: Galanz Enterprises Group Co. Ltd.

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